Printed Pages – 6 Roll No. : .....

# D020712(020)

## B. Tech. (Seventh Semester) Examination, Nov.-Dec. 2022

**(New Scheme)**

**(Civil Engg. Branch)**

## DESIGN of HYDRAULIC STRUCTURES

***Time Allowed : Four hours***

***Maximum Marks : 100***

***Minimum Pass Marks : 35***

*Note : Part (a) of each question is compulsory. Attempt any **one** from (b) and (c). Distribution of marks : Part (a) 4 marks, and part (b) and (c) 16 marks each.*

**Unit-I**

1. (a) What is wave pressure in a gravity dam?

   (b) (i) Explain the method of determining principal and shear stresses in a gravity dam.

   (ii) What are the various modes of failure of a gravity dam? Discuss each of them.

(c) Figure shows the section of a gravity dam built of concrete. Calculate (neglecting earthquake effects)

(i) The maximum vertical stresses at the heel and toe of the dam.

(ii) The major principal stress at the toe of the dam.

(iii) The intensity of shear stress on a horizontal plane near the toe.

Assume weight of concrete = 23.5 kN/m$^3$; and unit length of dam. Allowable stress in concrete may be taken 2500 kN/m$^3$.

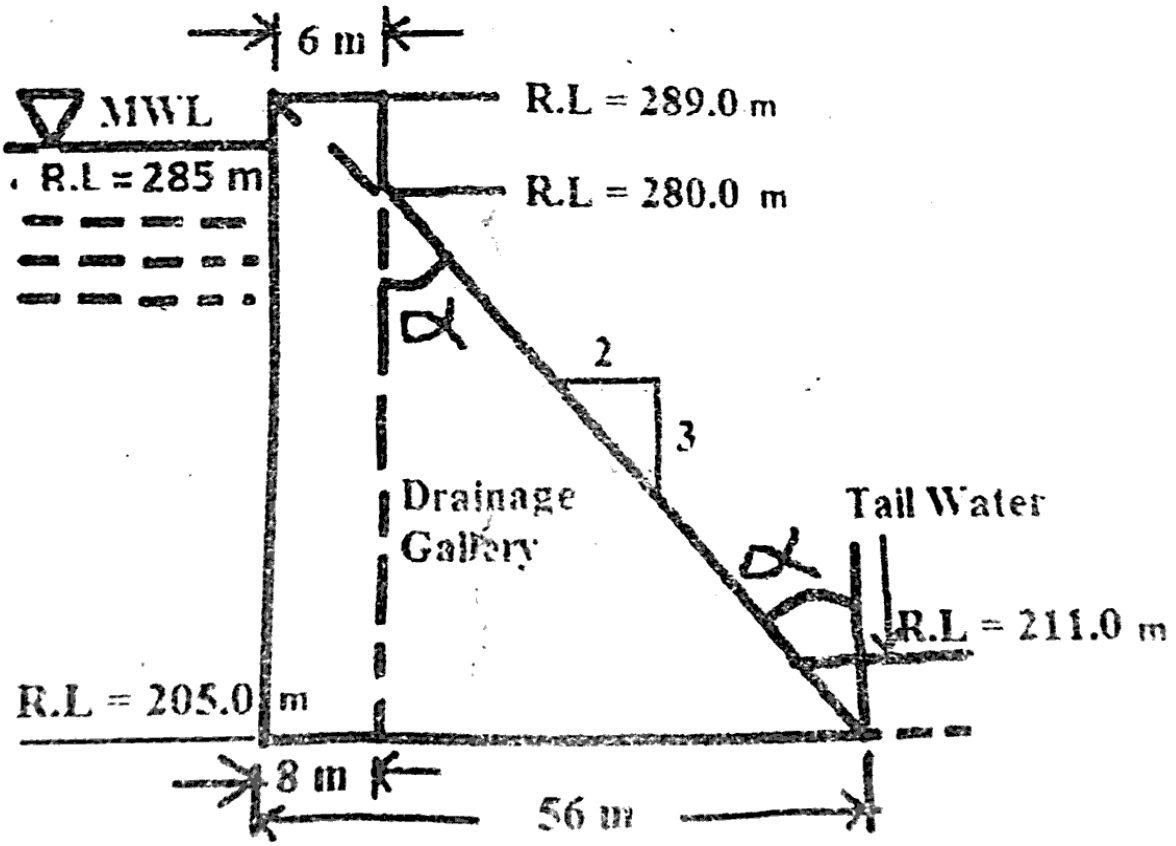


Fig. 1 (c)

## Unit-II

2. (a) Define high and low ogee spillway.

(b) Discuss the various types of energy dissipater used below spilway in relation to the poisition of the tail water curve and jump height curve.

(c) Design an ogee spillway for concrete gravity dam, for the following data :

(i) Average river bed level = 250.0 m

(ii) R.L. of spilway crest = 350.0 m

(iii) Slope of d/s face of gravity dam = 0.75:1

(iv) Design discharge = 6500 cumecs

(v) Length of spillway = 5 spans with a clear length of 9 m each

(vi) Thickness of each pier = 2 m.

## Unit-III

(a) What is under sluices?

(b) (i) Discuss on the effect of construction of weir on the river regime.

(ii) What is Bligh's creep theory? How does the Khosla's theory differ from Bligh's creep theory?

(c) A barrage is to be constructed on a river having flood discharge of about 8,100 cumecs, with a given data as follows :

Average bed level of river = 257.0 m

HFL (before construction of barrage) = 262.2 m

Permissible afflux = 1.0 m

Pond level = 260.6 m

Prepare a complete hydraulic design for the under sluice section as well as for other barrage bay. A safe exit gradient of 1/6 may be assumed 0.5 m. Retrogression and 20% discharge concentration.

## Unit-IV

4. (a) What is Regulation Work?

(b) (i) Describe the procedure of designing Sarda type fall.

(ii) Explain briefly Khosla's exit gradient concept.

(c) Design a straight flumed meter glacis fall with the following data :

Full supply discharge of the canal = 120 cumecs

Bed level of the canal upstream = 107.5 m

Bed level of the canal downstream = 106.0 m

Drop ($H_L$) = 1.5 m

FSL of the canal upstream = 109.7 m

FSL of canal downstream = 108.2 m

Bed width of the canale u/s and d/s = 60 m

Safe Exit Gradient for canal material = 1/5.5

## Unit-V

5. (a) What are the important features of design of CD works?

(b) (i) Write short note on :

1. Syphon aqueduct
2. Super passage
3. Canal syphon

(ii) What are the design considerations for aqueducts?

(c) Design a suitable cross-drainage works given the following data at the crossing of a canal and a drainage :

Canal

Full supply discharge = 32 cumecs

Full supply level = 213.5 R.L.

Canal bed level = 212.0 R.L.

Canal bed width = 20

Trapezoidal canal section with 1½ H : 1V slopes.

Canal water depth = 1.5 m

**Drainage**

High flood discharge = 300 cumecs

High flood level = 210.0 m

High flood depth = 2.5 m

General ground level = 212.5 m